STAINLESS STEEL PRODUCTS

In using these data it should be understood that they are results of laboratoty tests and are indicative only of the conditions under which they are run, and may be considered as a basis for recommendation, but not for guarantee. As actual operating conditions can rarely be duplicated in the laboratoty it is desirable to test further under actual service. The letters A, B, C, etc., represent the approximate corrosion penetration per month as follows:

CORROSION RESISTANCE DATA

substance and condition	temp F	304	316
Acetic acid			
5% agitated	70º	A	A
5% aerated	70º	A	A
10% agitated	70º	A	A
10% aerated	70º	A	A
20% agitated	70º	A	A
20% aerated	70º	A	A
50%	70º	A	A
50%	boiling	C	B
80%	70º	A	A
80%	boiling	D	B
100%	70º	A	A
100%	boiling	C	B
100%-150 lb. pressure	400º	E	C
acetone	boiling	A	A
	70º	A	A
acetylene	70º		
concentrated	70º	A	
commercially pure	70º		
alcohol, ethyl	70º	A	A
	boiling	A	A
alcohol, methyl	70º	A	A
	150º	C1	B
alum (chrome) 5%	70º	A	A
aluminium chloride	70º	D	C
aluminium	molten	E	E
aluminium sulphate			
10%	70º	A	A
	boiling	B	A
saturated	70º	A	A
	boiling	B	A
ammonia			
all concentrations	70º	A	A
gas	hot	D	
ammonia liquor	70º	A	A
	boiling	A	A
ammonium chloride			
1% still	70º	A	A
aerated	70º	A	A
agitated	70º	A	A
10% solution	boiling	A1	A1
28% solution	boiling	B1	A1
50% solution	boiling	B1	A1
ammonium nirate			
all concentrations			
agitated	70º	A	A
aerated	70º	A	A
saturated	boiling	A	A
ammonium phosphate 5%	70º	A	A
ammonium sulphate			
1% & 5% agitated	70º	A	A
aerated	70º	A	A
10%	boiling	B1	A1
saturated	boiling	B1	A1
ammonium sulphite	cold	A	A
	boiling	A	A
aniline 3%	70º	A	A
concentrated crude	70º	A	A
barium chloride 5%	70º	A	A
saturated	70º	A	A
aqueous solution	hot	B1	A1
barium nitrate			
aqueous solution	hot	A	A
barium sulphate			
barytes-blanc fixe	70º	A	A
barium sulphide			
saturated solution	70º	A	
solution	70º		A

substance and condition	temp F	304	316
beer		A	A
benzene	70º	A	A
benzol	hot	A	A
blood (meat juices)	cold	A1	A
boric acid			
5% solution	70º	A	A
5% solution hot			
5% solution boiling			A1
saturated solution	70º	A1	A1
boilding			A1
buttermilk	70º	A	A
Calcium carbonate	70º	A	A
calcium chlorate			
dilute solution	70º	A	A
dilute solution	hot	A	A
calcium chloride			
dilute solution	70º	B2	A1
concentrated solution	70º	B2	A1
calcium hydroxide			
10%	boiling	A	A
20%	boiling	A	A
50%	boiling	C	B
calcium sulphate			
saturated	70º	A	A
carbonated water		A	A
carbonic acid			
saturated solution	70º	A	
carbon monoxide gas	870ºC.	A	A
	760ºC.	A	A
carbon tetrachloride			
pure	70º	A	A
aqueous solution 5-10%	70º	C1	
chloracetic acid	70º	D	C
chloric acid	70º	E	D
chlorinated water			
saturated	70º	C1	B1
chlorine gas - dry	70º	C	B
gas - moist	70º	D	C
gas	100ºC.	E	D
chloroform	70º	A	A
chromic acid			
5%	70º	A	A
10% CP	boiling	C	B
50% commerical (cont. SO2)	boiling	D1	C
cider	70º	A	A
citric acid			
5% still	70º	A	A
5% still	150º	A	A
15%	70º	A	A
15%	boiling	B	A
concentrated	boiling	C	B
coca cola syrup (pure)	70º	A	A
coffee	boiling	A	A
copper carbonate			
sat. sol. in 50% NH_3OH		A	A
copper chloride			
1% agitated	70º	B1	A1
1% aerated	70º	B1	A1
5% agitated	70º	C1	B1
5% aerated	70º	E1	D1
copper cyanide			
saturated solution	boiling	A	A
copper nitrate			
1% still	70º	A	A
1% agitated	70º	A	A
1% aerated	70º	A	A
5% still	70º	A	A

substance and condition	temp F	304	316
5% agitated	70º	A	A
5% aerated	70º	A	A
50% aqueous solution	hot	A	A
copper sulphate			
5% agitated still	70º	A	A
5% aerated	70º	A	A
saturated solution	boiling	A	A
creosote (coal tar)	hot	A	A
creosote oil	hot	A	A
dyewood liquor	70º	A3	A
ether	70º	A	A
ethyl acetate			
concentrated solution	70º	A	
ethyl chloride	70º	A	A
ethylene chloride	70º	A	A
ethylene giycol			
concentrated	70º	A	
Ferric chloride			
1% solution still	70º	B41	A1
	boiling	D41	C1
5% solution still	70º	D431	C1
5% agitated	70º	C431	C1
5% aerated	70º	C431	C1
ferric hydroxide			
(hydrated iron oxide)	70º	A	A
ferric nitrate			
1% still	70º	A	A
5% still	70º	A	A
1% agitated	70º	A	A
5% agitated	70º	A	A
1% aerated	70º	A	A
5% aerated	70º	A	A
ferric sulphate			
1% and 5% still	70º	A1	A
aerated or agitated	70º	A1	A
ferrous chloride			
saturated solution	70º		A
ferrous sulphate			
dilute solution	70º	A	A
fluorine	70º	E	E
formaldehyde			
40% solution		A1	A1
formic acid			
5% still	70º	B	A
	150º	B	A
fruit juices	70º	A	A
fuel oil	hot	A	A
containing sulphuric		C	B
gasoline	70º	A	A
gelatine			A
gule - dry	70º	A	A
solution - acid	70º	B1	A
	140º	B1	A
glycerine	70º	A	A
Hydrachloric acid			
all concentrations	70º	E	E
hydrocyanic acid		A	A
hydrofluosilicic acid	70º	E	D
hydrogen sulphide			
dry		A	A
wet		B3	A3
Ink		B3	A
iodine		E	D
iodoform		A	A
Kerosene	70º	A	A
ketchup	70º	A1	A

1 Subject to pitting at air line or when allowed to dry.
2 Keep solutions alkaline.